



JOURNAL OF QUALITY ASSURANCE IN AGRICULTURAL SCIENCES (JQAAS)

Impact of Weather Factors on the Population Dynamics of Major Sucking Insect Pests of Okra in Tandojam

Mukhtar Ahmed¹ and Jan Muhammad Marri^{1*}

¹Department of Plant Protection, Sindh Agriculture University Tandojam

*Corresponding Authors:

Article Received 12-04-2026, Article Revised 20-06-2026, Article Accepted 20-08-2026

ABSTRACT

To know the seasonal population of sucking insect pests of okra with its relationship of the ecological components at surroundings of Tandojam, a field study was conducted at the experimental area at Horticulture Section, Tandojam during 2019 under the title of “Impact of weather parameters on the population pattern of sucking insect pests okra in Tandojam”. The sucking insect species *Bemisia tabaci*, *Thrips tabaci*, *Aphis gossypii* and *Aphis bigutella bigutella* were observed on okra crop in the 3rd week of February till June with an increase in their population during the month of April on this variety of okra. Additional data on overall seasonal average population of sucking insect complex on okra revealed that the average numbers of whitefly, jassid, thrip and aphid recorded on okra crop during the cropping season were (15.87 ± 2.45) , (21.50 ± 3.50) , (14.50 ± 2.56) and (13.45 ± 2.78) , respectively. Further environmental factors play important role on insect reproduction, population build up and other life activities. As result showed that ecological parameters had positive and negative impact on sucking insect pest population on okra crop. However, temperature showed significant positive with whitefly ($r = 0.67^*$), and highly positive with Jassid ($r = 0.89^{**}$), with thrip ($r = 0.55^*$) and with aphid ($r = 0.49^*$), respectively at ($P < 0.05^*$ and $P < 0.01^{**}$). Furthermore, relative humidity showed negative correlation such as, with whitefly ($-r = 0.55^*$), with Jassid ($-r = 0.33^{**}$), with thrip ($-r = 0.61^*$) and with aphid ($-r = 0.63^*$), respectively at statistically level ($P < 0.05^*$ and $P < 0.01^{**}$), respectively.

Keywords: Sucking insect pests, Okra (*Abelmoschus esculentus* L.), Climatic factors

INTRODUCTION

Commonly increasing population and changes in dietary habits related with urbanization increased the demands for food. Pakistan has become world's largest producing and consuming the vegetables (FAO, 2019). Among them Okra, *Abelmoschus esculentus* (L.) is also known as lady finger and one of the most popular, commercially, significantly grown worldwide and in the tropical and subtropical areas for its immature edible green fruits (George, 1999). In Pakistan it is cultivated in nearly all provinces, recently data shows that in Pakistan cultivation of okra crop was around 30 percent of okra for fulfill the requirement for domestic consumption. The total area under okra cultivation during 2016-17 was 279 thousand tons thousand tons; while the seed production was thousand 61 tons, respectively. Okra or lady finger mainly is hot seasonal crop which belongs to the family Malvaceae to be initially known the originate from African countries. It is the rich source of vitamins, minerals like Fe, Mn, Zn, Ni (Bhatt *et al.*, 2018). Medicinally, okra is useful in spermatorrhea, genito-urinary disorder and chronic dysentery. It also cures ulcers and provides relief from

hemorrhoids (Gemedé *et al.*, 2014). Okra is a vegetables, which are devoted almost on a each day basis in many family and for that reason it is considered one of essential components of the diet (El-Shafie, 2001; FAO, 2008). It is a short period vegetable crop cultivated and some farmers commonly grown other crops (Norman, 1992), which practice almost globally exercised. There was reduction in okra production; and apart from the reduction in area under okra cultivation, the lower production was also associated with the insect pest infestation, particularly the sucking complex (Parveen and Dhandapani, 2002). Insect pests constitute an important constraint in the successful cultivation of the crop (Bhatt and Karnatak, 2018). The yield of okra is drastically hampered by the attack of various pests like shoot and fruit borer, American bollworm or fruit borer, whitefly, leafhopper, aphids, leaf roller, blister beetle, dusky cotton bug, red spider mite (Kumar *et al.*, 2017). The key insect-pests causing the economic losses in okra are: leaf hopper, fruit borer, whitefly, thrips and shoot and fruit borer (Solangi *et al.*, 2008). However, the level of infestation and the losses caused by these pests vary from region to region as well as the season in which the crop is grown. Broadly, their

infestation depends upon the environmental conditions i.e. temperature, rainfall, wind speed and relative humidity either positively or negatively. Environmental conditions had an important influence on the density of aphid, jassid and other sucking insect pests. Maximum and minimum temperature had a major impact on the negatively of mustard aphid when correlated with environmental conditions. Environmental conditions such as sunshine, evaporation, rainfall and maximum relative humidity were also exhibited a negatively correlation with aphid multiplication. The reason for this negative correlation could possibly be due to the fact that population of aphid did not synchronize with respective parameters. Highest aphid population was obtained at 18-20°C. A significant correlation was also obtained between mean temperature and the number of pest populations. The number of abundant individuals fluctuated from year to year, depending primarily on the rainfall and the temperature conditions. The environmental factors play essential functions in terms relating to achievement and letdown the agriculture crops, because of such components powerfully persuade the physiological appearance of hereditary prospective of the crops. It has been well reported and documented that the yield or of various development stages of crop is strongly depend on the weather conditions, such as relative humidity, temperature, soil moisture and solar radiation and other several factors. Addressing to the above issue, consequently current research may be conducted to evaluate the influence of environmental influences on the population inclination of sucking insect pests in okra crop in Tandojam vicinity. The present study not only helps to understand the role of weather parameters in the sucking complex pests of okra crop. But it will also provide the essential information, which will help assistance to develop the potential pest control methods.

OBJECTIVES:

1. To determine the seasonal population trend of sucking insect pest in okra crop.
2. To evaluate the impact of weather parameters on the population of sucking complex and damage extent caused by sucking insect pests.

MATERIALS AND METHODS

Trial site and period: A field experiment entitled “Influence of weather parameters on the population pattern of sucking insect pests okra in Tandojam” was conducted at the experimental area of Horticulture Section, Tandojam during 2019, to asses the seasonal population of sucking insects with the relation to the ecological aspects at Tandojam surroundings.

Experimental set up: The Sabzpari variety was planted in the area of experimental plots total size half acre for the investigation of population of sucking complex with three sub-plots size 15 × 15 meters was further divided with three replications. The experiment was designed on Randomized Complete Block Design

(RCBD), crop was sown on second week of February 2019. Recommended agronomical practices i.e land preparation, seed sowing time, seed bed preparation number of irrigation, weeding and manure application were done. Further plant sequence practices were also made such as space row to row 75 cm and plant to plant will be 25 cm. No any effective measure was taken the against the population buildup of *A. gossypii*, that may be effect the natural population condition in a specific season.

Data on population of sucking complex on okra crop: For data observation and record on the seasonal population of sucking insect pests such as aphid, whitefly, jassid and thrip on okra vegetable crop was recorded; in total fifty plants was selected randomly from each sub plot. The population trend of each insect species was recorded and observed from 03 leaves from each plant one from bottom, one from middle and one from top leaves were inspected. Population trend of sucking insects pests was observed after each week duration during crop season.

Damage Based on leave symptoms observation and recorded: The assessment of relative damage percentage caused by sucking insects were examined and recorded by the affected leaves based on the standard and well reported or documented symptoms of sucking complex in okra crop . The damaged leaves were observed by the sign which carried by those insect pests. The total one hundred okra plant and six leaves per plant (two from top, two from middle and two from down) were examined and then damaged percentage was ascertained.

Weather parameter: The abiotic factor data was collected (Temp. and R.H) at the weekly intervals at in the field of experimental area. Whole year environmental data was obtained from meteorological department ARI, Tandojam.

Statistical analysis: The impact of various weather conditions or abiotic factor parameters like average temperature and average relative humidity were recorded to determine the impact on population trend of sucking insect pests on okra crop were investigated through correlation studies, To examine the consequence of weather factors on sucking insect pests population was correlated using same statistical process. The LSD teat was performed through statistical software 8.1.

RESULTS

A field study entitled “Impact of weather parameters on the population pattern of sucking insect pests okra in Tandojam” was carried out at the experimental area of Horticulture Section, Tandojam during 2019, to investigate the seasonal population of sucking insects with the relation to the ecological components at Tandojam surroundings. For this purpose Sabzpari variety was cultivated during the second week of Fabuary-2019 and after one month, the data on the population of sucking complex such as white fly *Bemisia tabaci*, Thrip, *T. tabaci*, Aphid, *A.*

gossypii and Jassid *Amrasca biguttula biguttula* were as observed and recorded till the end of crop. The main results of the present experiment are described as under:

Population of white fly *Bemisia tabaci* : Figure 01 indicated that the population of *B. tabaci* was found at the initial stage of the okra crop throughout the cropping season. The data revealed that the initially whitefly population (5.58/per leaf) was recorded on

March 12, 2019 on okra and gradually increased its population later. Further, sever population of *B. tabaci* was recorded during the last week of April, that is the numbers of whitefly were (17.40/per leaf) and next weeks the population was fluctuated and the pattern was varied. However, Minimum population was observed in the first second week that is (5.50) and second minimum was observed third week of June-2019 (8.60/per leaf), respectively

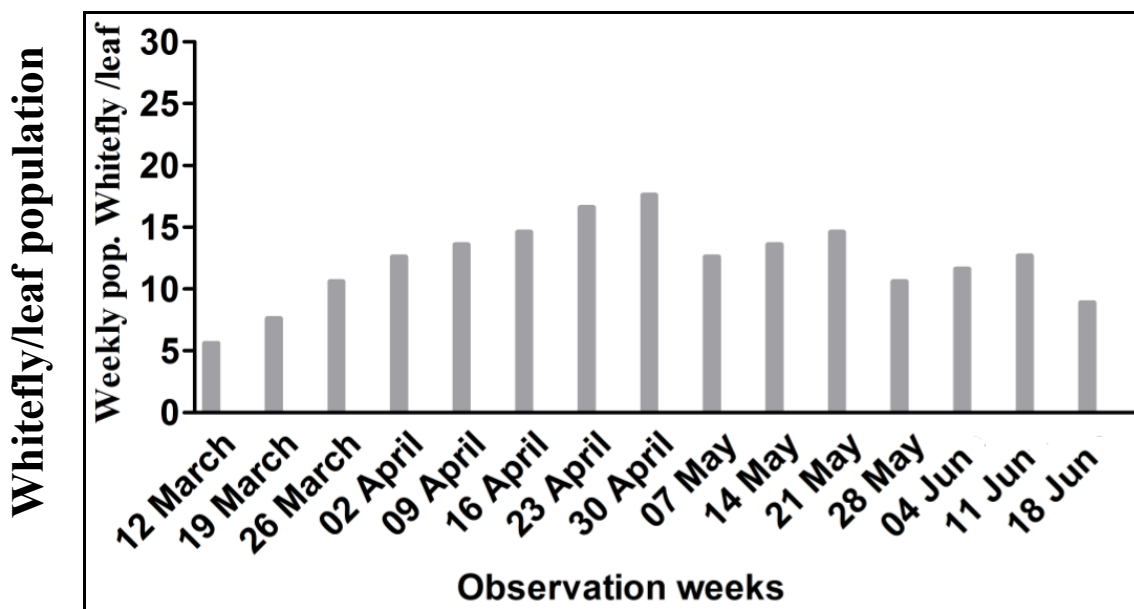


Figure 1. Weekly seasonal populations of White fly per leaf on okra crop during 2019.

Population of Jassid *Amrasca biguttula biguttula*: The data indicated jassid was appeared in near the beginning of crop and population of jassid was observed through out the cropping seasons and infestation was also observed in okra crop. The data presented (Figure 2) revealed that the at the start jassid population (9.40/per leaf) was recorded on March 12, 2019 on okra and gradually increased it's population. Further, peak population of *A. biguttula biguttula* was

recorded during the 3rd week of April, that is the numbers of jassids were (29.30/per leaf) and next weeks the population was decreased and the pattenen was varied. However second heights number of Jassid were noted 2nd week of june-2019, which was (27.80/per leaf). However, minimum population was observed in the first second week that is (9.40) and second minimum was observed 3rd week of March 2019 (13.60/per leaf), respectively.

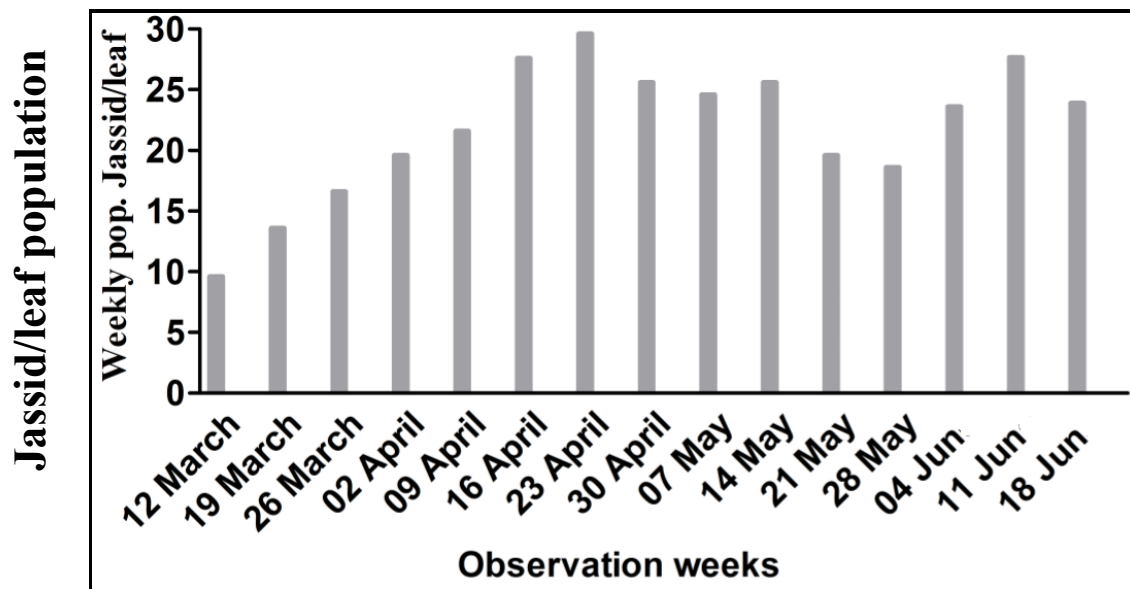


Figure 2. Weekly seasonal populations of Jassid per leaf on okra crop during 2019.

Population of Thrip, *Thrips tabaci*: The data indicated thrip was seemed in the beginning of crop and population of thrip was observed through out the cropping seasons and infestation was also observed in okra crop. The data presented (Figure 3) revealed that the at the start of thrip population (7.30/per leaf) was recorded during the first week of observation such as on March 12, 2019 on okra and gradually increased it's population. Further maximum number of thrip

population was observed during the 3rd week of May, that is the numbers of Thrip (25.25/per leaf) and next weeks the population was decreased and the pattern was varied. However second heightst number of thrips were noted 2nd week of April which was (21.80/per leaf). However, minmium population was observed in the first second week that is (7.40) and second minimum was observed 2nd week of March 2019 (11.60/per leaf), respectively.

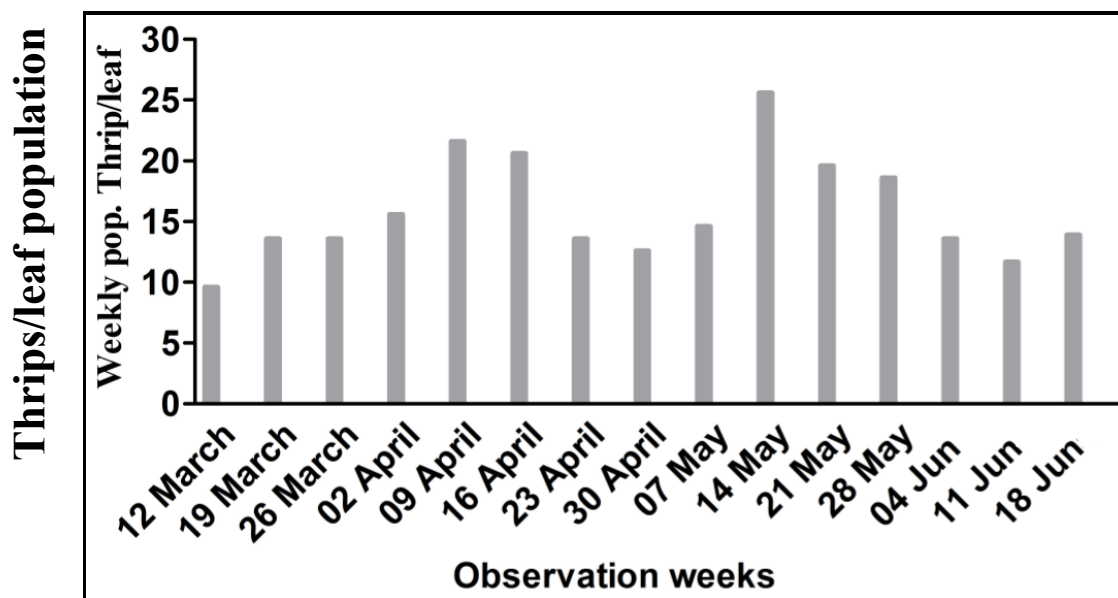


Figure 3. Weekly seasonal populations of thrip per leaf on okra crop during 2019.

Population of Aphid, *Aphis gossypii*: The data indicated aphid population was appeared in the early stage of okra crop and population of thrip was observed through out the cropping seasons. The data presented (Figure 4) revealed that the at the start of thrip population (6.60/per leaf) was recorded during the first week of observation such as on March 12, 2019 on okra and gradually increased it's population. Further

maximum number of thrip population was observed during last week of April, that is the numbers of Thrip (21.40/per leaf) and next weeks the population was decreased toward end of crop. However second heightst number of aphid were 4th week of April which was (20.80/per leaf). However, minimum population was observed in the first second week that is (6.60) and

second minimum was observed 2nd week of June 2019 (7.60/per leaf), respectively.

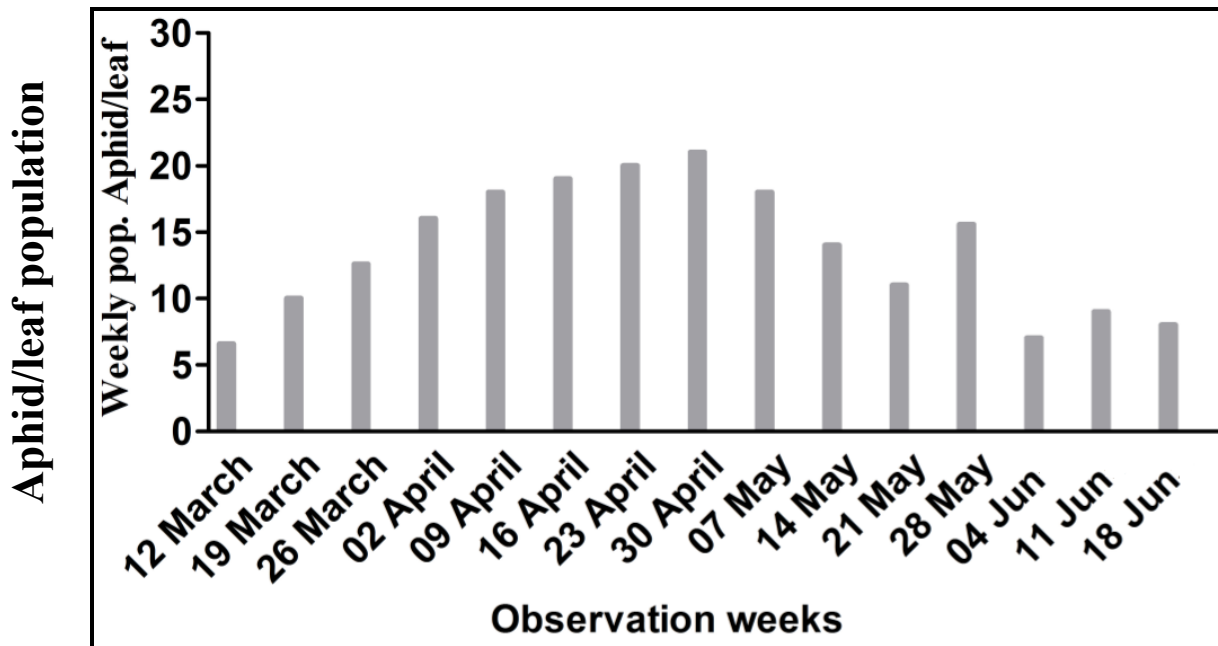


Figure 4. Weekly seasonal populations of aphid per leaf on okra crop during 2019.

Further results in (Figure 5) revealed that overall seasonal average population of sucking insect complex on okra during 2019. As population observed

(15.87±2.45), Jassid (21.50±3.50), Thrip (14.50±2.56) and Aphid (13.45±2.78) were recorded as depicted in (Figure 5), on okra crop during cropping season, respectively.

and recorded on okra crop and weekly data has already described in on individual basis in (Figures 1-4). Furthermore, mean population of whitefly was

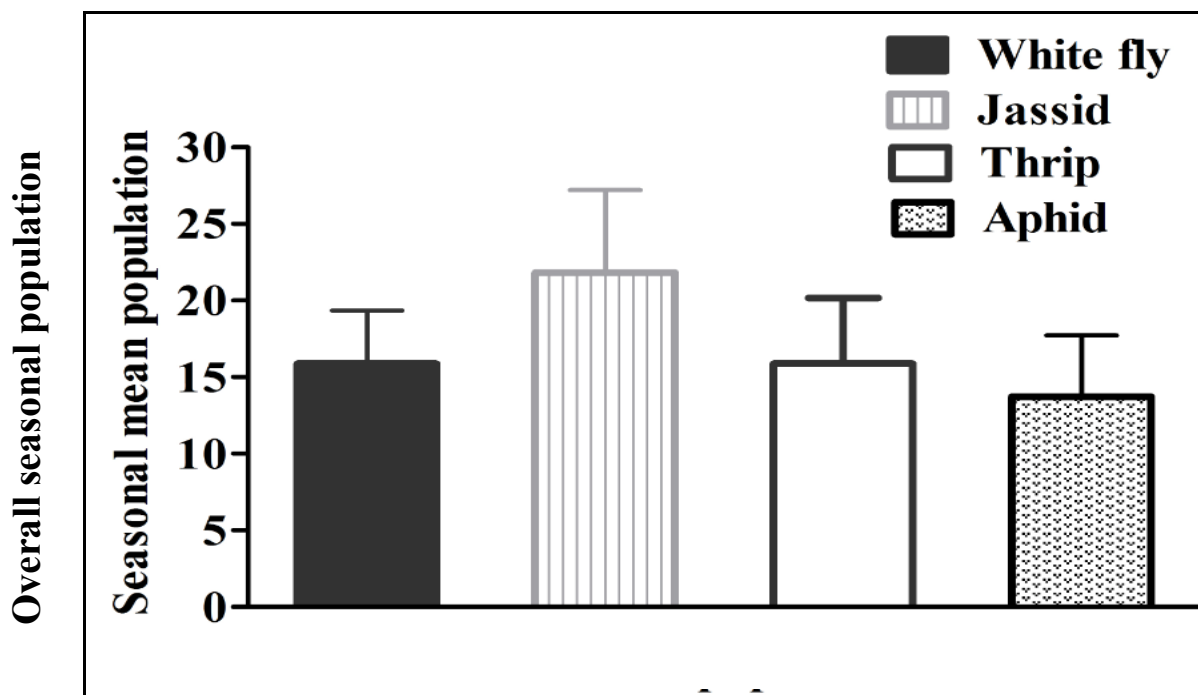


Figure 5. Overall seasonal (means ± SE) populations of sucking insect pests per leaf on okra crop during 2019.

Further environmental factors play important role on insect reproduction, population build up and other life

activities. As result in (Table 1) showed that ecological parameters had positive and negative impact on sucking

insect pest population on okra crop. However, temperature showed significant positive with whitefly ($r = 0.67$), and highly positive with Jassid ($r = 0.89$), with thrip ($r = 0.55$) and with aphid ($r = 0.49$), respectively at ($P < 0.05$ and $P < 0.01$). Furthermore, relative humidity

showed negative correlation such as, with whitefly ($r = 0.55$), with Jassid ($r = 0.33$), with thrip ($r = 0.61$) and with aphid ($r = 0.63$), respectively at statistically level ($P < 0.05$ and $P < 0.01$), respectively.

Table 1: Effect of environmental conditions on abundance of sucking insect pests on okra during 2019 at Tandojam surroundings.

Insects	Relative Humidity % (r)	Temperature° C (r)
White fly	-0.55*	0.67*
Jassid	-0.33**	0.89**
Thrip	-0.61*	0.55*
Aphid	-0.63*	0.49*

** = Highly significant

* = Significant

DISCUSSION

Okra, *Abelmoschus esculentus* usually famous and known as *bhindi* or lady's finger associated with the the family Malvaceae which is a essential vegetable crops and supposed to be initiated grown from Africa. It is a significant tropical, sub-tropical seasonal vegetable cultivated almost all parts of the world. It is developed in an area of 551.64 thousand hectares with an annual production 31.39 thousand metric tonnes and productivity of 51.51 q/ha (Anonymous, 2017-18). It is one of the essential money crops paying high returns to the farmers. The crop is attacked near on the subject of hundred species of insect pests right from sowing to harvest (Rao and Rajendran, 2003). Among them sucking insect pests are leafhopper, *Amrasca biguttula biguttula*, whitefly, *Bemisia tabaci*, aphid, *Aphis gossypii* and mite, *Tetranychus cinnabarinus* (Meena and Kanwat, 2005; Kumar et al., 2014). A field experimental entitled "Influence of weather parameters on the population pattern of sucking insect pest's okra in Tandojam" was carried out to determine the seasonal population of sucking insects with the relation to the ecological parameters at Tandojam surroundings. The major sucking insect species such as *B. tabaci*, *T. tabaci*, *A. gossypii* and *A. bigutella bigutella* were recorded on okra crop just three weeks after sowing till June. The results revealed that the initially whitefly population was appeared on March and maximum whitefly population was recorded in the in the month of April. The our results reported are in agreement with those of Patel and Jhala, (1992) who observed that from March to May were the most favorable months for the reproduction and survival of white flies. The results are well supported by Dutta and Dutta (1997) who reported white fly a as the most important pest of Malvaceae family that commonly occurred on okras. The population of whitefly behaved positive response with temperature and significant negative with relative humidity, further Singh and Sachan (1997) recorded that white fly showed a positive correlation with temperature and and negative with relative humidity. Initially population of Jassid *A. biguttula biguttula* was appeared from mid-March till month of June 2019. In

the same way peak population of jassid was recorded during the third week of April and second and third week of June. Previous studies have also reported that *A. biguttula biguttula* appeared in mid-March and reached to highest in month of April (Abro et al., 2004; Solangi et al., 2008). Fuether, Hanumantharaya et al. (2008) and Godhani et al. (2009) had reported population frequency of Jassid was recorded throughout the cropping season, this, evidences were highly simila to the results of present study. Furthermore, population trend of okra Jassid *A. biguttula biguttula* showed significant positive response with temperature and negative with relative humidity in okra crop in Tandojam vicinity. Further, findings of Godhani et al. (2009) closely related with present research the frequency of *A. gossypii* throughout the cropping development time. In the current research the population of *A. gossypii* reduced from March-May. Results of our study closely related with the findings of Abro et al. (2004) and Solangi et al. (2008). They had stated that the average maximum population of Jassid was appeared at the end of April. The current research indicated that the R.H presented negative impact on okra aphid population but temperature influence positively. Studies further studies revealed that in thrip population was appeared also in the month of March and highest population was observed in the month of April and this population of thrip was buildup and fluctuated throughout cropping seasons. The population of thrip was below Economic threshold level (ETL) from March to onward. The result of the present study are very much agree with the results of Khan and Ullah (1994), they confirmed a negative relationship between population buildup of *T. tabaci* and the mean relative humidity and positive relationship with temperature. The current results are very close with the results obtained by Linger et al. (1998) and Sadras and Wilson (1997) who reported that aphid population was favored by higher temperature.

CONCLUSIONS

Four major sucking insect pest species were noticed to attack the okra crop from sowing to

harvesting. Further, on the basis of results all sucking insect pests were appeared in early stage of crop till maturity and population of were found through out the cropping seasons and infestation were observed in okra crop. Further enviornmental factors play important role on insect reproduction, population build up and other life activities and also playing an important role on the behavior of insect pests in okra crop, however, all sucking complex showed positive relationship with temperature and negative with relative humididy

RECOMMENDATIONS

Regular pest couting should be done in okra field to prevent the insect pest damages.

Chemical control tactics on fortnight basis should be used to eradicate the sucking complex problem in okra field.

Integrated pest management approaches may be adopted to save the okra crop from these noxious insects pests.

REFERENCES

- Abdalla, A. Satti 2012. The major predators and their seasonal abundances in okra field of Et al Gorair scheme northern Sudan. J. Plant Sci., 4 (2): 271-276.
- Abro, G.H., T.S. Syed, G.M. Tunio and M.A. Khuhro. 2004. Performance of transgenic Bt cotton against insect pest infestation. J. Biotech., 3(1): 75-81.
- Anonymous. Okra: Area under cultivation. National Horticulture Board, 2017-18. Website: <http://nhb.govt.in/bulletin-vegetables.html>.
- Ashfaq, S., I.A. Khan, A.U.R. Saljoqi, S. Ahmad, F. Manzoor, K. Sohail, K. Habib and A. Sadozai. 2011. Population dynamics of insect pests of cotton and their natural enemies. Sarhad J. Agric., 2 (2): 251-253.
- Bhatt, B. and A. K. Karnatak. 2018. Population dynamics of sucking pests and their predators on okra agroecosystem. J. Entom. and Zool. Stud., 2 (1): 1289-1291.
- Dutta, S. K and B. C Dutta. 1997. A preliminary survey of insect pests of rapeseed-mustard in Central Brahmaputra valley zone of Assam. J. Plant Heal., 1 (15): 20.
- El-Shafie, H. A. F. 2001. The use of neem products for sustainable management of homopterous key pests on potatoes and eggplants in the Sudan. Ph.D thesis. Institute of Phytopathology and Applied Zoology. Justus Liebig University of Giessen.
- F.A.O. 2000. Eggplant Ecological Guide. FAO Inter-Country Programme for IPM in Vegetables in South and Southeast.
- F.A.O. 2008. Small farmer participation in export production: The case of Kenya.
- F.A.O. 2019. Cabbage Ecological Guide. FAO Inter-Country Programme for IPM in Vegetables in South and Southeast Asia. Rome.
- Gemed, H.F., N. Ratta, G. D. Haki, A. Z. Woldegiorgis and F. Beyene. 2014. Nutritional quality and health benefits of okra (*Abelmoschus esculentus*). J. Food Sci. and Qual. Manag., 33 (2): 87-96
- George, R.A.T. 1999. Vegetable seed population (2nd edition), CABI Publication, 328 pp.
- Godhani, P.H., R.M. Patel, J.J. Jani, D.N. Yadav, D.M. Korat and B.H. Patel. 2009. Impact of habitat manipulation on insect pests and their natural enemies in hybrid cotton. Karnataka J. Agric. Sci., 22(1): 104-107.
- Gulati, R. 2004. Incidence of Tetranychus cinnabarinus (Boisd.) Infestation in different varieties of Abelmoschus esculentus L. Annals of Pl. Prot. Sci., 12: 45-47.
- Hanumantharaya, L., K.B. Goud and L.K. Naik. 2008. Use of green lacewing, *chrysoperla carnea* (Stephens) and Neem seed kernel extract for management of insect pests on cotton. Karnataka J. Agric. Sci., 21(1): 41-44.
- Khan, S. M and Z. Ullah. 1994. Population dynamics of insect pests of cotton in Dera Ismail Khan. Sarhad J. Agric., 10 (1):285-290.
- Kumar U, Singh DV, Sachan SK, Bhatnagar A, Singh R. 2014. Insect pest complex of okra and biology of shoot and fruit borer. Ind. J. Entom., 76 (1): 29-31.
- Kumar, P., Singh, D.V. and Sanchan, S.K. 2017. Succession of Important Insect Pests of Okra *Abelmoschus esculentus* Under Western Uttar Pradesh Climatic Conditions. Int. J. Eng. and Manag. Sci., 1 (1): 72-74.
- Lanjar, A. G. and H. A. Sahito 2007. Impact of weeding on whitefly, *Bemisia tabaci* (genn.) population on okra crop. Pak. J. Weed Sci. Res., 13 (1): 209-217.
- Linger, M. S., J. S. Awaknavar and S. Lingapa. 1998. Seasonal occurrence of chilli mites and thrips. J. Agric. Res., 11 (2): 380-385.
- Mari, J. M. 2013. Impact of trap crops on the population of jassid (Amrasca devastans (dist.) in okra. J. Agric. Res., 9 (1): 252-255.
- Meena, N. K and P.M. Kanwat. 2005. Broad reconnaissance of insect pests of okra in semi-arid region of Rajasthan. J. Appl. Entom., 55 (2): 261-262.
- Miguel, A. A. Deborah K. L. and Stephen J. R. 2008. Vegetation diversity and insect pest outbreaks. 2:131-169.
- Miguel, A.A. and C.I. Nicholls 2003. Soil fertility management and insect pests: harmonizing soil and plant health in agro-ecosystems. J. Soil and Tillage Res., 2 (1): 203-211.
- Munthali and Tshegofatso 2013. Major insects of Okra in Sebele. J. Agric. Appl. Sci., 2: (1): 234-245.
- Nath, P., O. P. Chaudhary, P. D., Sharma and H. D. Kaushik. 2000. The studies on the incidence of important insect pests of cotton with special reference to *Gossypium arborium* (Desi) cotton. Ind. J. Entom., 62(1): 391-395.
- Nisha, S. 2014. Seasonal incidence and relation to weather parameters of aphid and their natural enemies on okra. Int. J. Sci. and Res., 4 (1): 2250-3153.
- Norman, J.C. 1992. *Tropical vegetable crops*. Pub Arthur H. Stockwell Ltd. 252p.
- Patel, H. M. and R. C. Jhala. 1992. Studies on host range, host preference and population dynamics of whitefly, *B. tabaci* (Gennadius) in south Gujarat, India. Guj. Agri. Uni. Res. J., 17 (2): 76-81.
- Prakash, Y. S., P. B. S. Bhadoria and R. Amitava. 2002. Relative efficiency of organic manures in improving resistance and pest tolerance of okra. J. Ann. Agri. Res., 23 (1): 25-531.
- Praveen, P.M and N. Dhandapani. 2002. Eco-friendly management of major pests of okra (*Abelmoschus esculentus* (L.) Moench). J. Vegetable Crop Prod. 2: 3-11.
- Rao, N.S. and R. Rajendran 2003. Joint action potential of neem with other plant extracts against the leafhopper, *Amrasca devastans* (Distant) on okra. J. Pest Manag. and Econ. Zool., 10 (2):131-136.
- Sadras, V. O., L. J. Wilson. 1997. Growth analysis of cotton crop infested with spider mites. Crop Science 37:481-491.
- Sahito H.A., M. A. Talpur, M. A. Soomro, A. H. Mastoi and K. H. Dhilloo 2013. Feeding Efficacy of spiders on sucking complex of okra, *A. Esculentus* L. J. Agric. and Sustain., 2 (1): 142-159.
- Sarkar, P. K., A. B. Mukherjee and J. Ghosh. 1996. Assessment of loss of bhendi against red spider mite. J. Env. Eco., 2:480-481.
- Singh, C. P and F. C. Sachan. 1997. Estimation of losses in yield of rapeseed, (*Brassica campestris*), by the mustard aphid, (*Lipaphis erysimi*) (Kalt.) in Tarai, India. J. Insect Sci., 16 (3/4):283-286.

Publisher's note: JQAAS remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



This is an open access article distributed under the terms of the Creative Commons Attribution 4.0 International License (CC BY 4.0), which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. To

view a copy of this license, visit [Creative Commons Attribution 4.0 International License \(CC BY 4.0\)](https://creativecommons.org/licenses/by/4.0/)